SIMATS ENGINEERING

Assessment Tool 3

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Course Code : CSA4307

Course : Internet Programming

Scenario

A start-up is developing a modern web application. During development the team encounters issues related to HTML elements, HTTP methods, XML syntax and web communication protocols. The team needs to embed a promotional video on the homepage, send registration data securely to the server, implement a semantic navigation menu, correct XML configuration files that contain syntax errors, and use the correct web protocol.

Question 1

Select the correct HTML tag for video embedding and justify your answer.

Answer: Option B, <video src="video.mp4" controls></video>.

Justification: The <video> element is the HTML5 element designed to embed video in a web page. The src attribute gives the location of the file, and the controls attribute displays the browser's built-in play, pause, seek and volume interface. Options A (<media>), C (<movie>) and D (<vid>) are not HTML elements, so browsers ignore them and the video does not play.

Code (index.html):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Promotional Video</title>
</head>
<body>
  <h1>Welcome</h1>
  <video src="video.mp4" controls width="480"></video>
</body>
</html>
```



Question 2

Identify the appropriate HTTP method for form submission and explain why.

Answer: Option B, POST.

Justification: POST sends the form data in the body of the HTTP request, so the data does not appear in the URL, browser history or server access logs. This is required for registration, where personal data and a password are sent to the server. GET appends the form data to the URL as a query string, which exposes the password, as shown in Figure 3. PUT and DELETE are REST methods for replacing or removing a resource; HTML forms do not natively send them. POST does not encrypt data by itself, so the application should also be served over HTTPS.

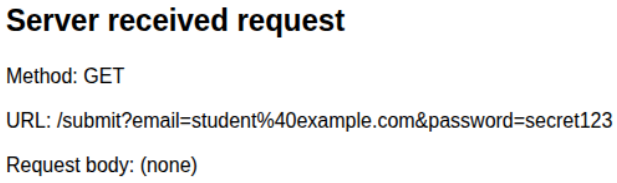
Code (register_post.html, correct method):

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Registration (POST)</title></head>
<body>
  <h2>Register</h2>
  <form action="/submit" method="POST">
    <p>Email: <input type="email" name="email"></p>
    <p>Password: <input type="password" name="password"></p>
    <button type="submit">Register</button>
  </form>
</body>
</html>
```

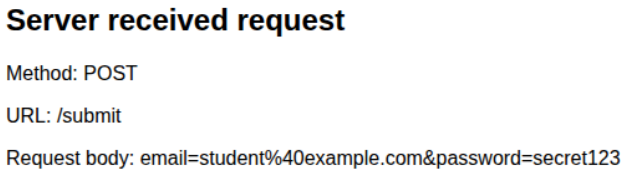
Code (register_get.html, incorrect method, for comparison):

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><title>Registration (GET)</title></head>
<body>
  <h2>Register</h2>
  <form action="/submit" method="GET">
    <p>Email: <input type="email" name="email"></p>
    <p>Password: <input type="password" name="password"></p>
    <button type="submit">Register</button>
```

```
  </form>
</body>
</html>
```



Form submitted with GET. The email and password appear in the URL.



Form submitted with POST. The data is in the request body and the URL is clean.

Question 3

Choose the proper semantic element for navigation and justify its usage.

Answer: Option B, <nav>.

Justification: The <nav> element marks a block of navigation links. Its meaning is understood by browsers, search engines and assistive technologies such as screen readers, which allow users to skip directly to the menu.

Option A ( ) is a generic container with no meaning.

Option C (<section>) groups thematic content within a page rather than navigation links.
Option D (<header>) holds introductory content such as a logo or title, and it may contain a <nav>, but it is not the navigation itself.

Code (nav.html):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Semantic Navigation</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 20px; }
    nav ul { list-style: none; padding: 0; margin: 0; }
    nav li { display: inline-block; margin-right: 20px; }
    nav a { text-decoration: none; color: #000; font-weight: bold; }
    nav { border-bottom: 1px solid #000; padding-bottom: 8px; }
  </style>
</head>
<body>
  <nav>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">Products</a></li>
      <li><a href="#">About</a></li>
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>
  <section>
    <h2>Our Products</h2>
    <p>Content of the page goes here.</p>
  </section>
</body>
</html>
```



Question 4

Identify the correct XML syntax and explain the error in the other options.

Answer: Option B, <note><to>User</to><from>Admin</from></note>.

Justification: An XML document must have every element closed by a matching end tag, and the tags must be correctly nested. Option B meets both rules. Each option was saved as an XML file and opened in the browser, which reports any well-formedness error (Figure 5).

- **Option A:** the <note> element is never closed, because </note> is missing.
- **Option C:** the <to> element is closed with </from>. This mismatched end tag breaks the nesting, so the document is not well-formed.
- **Option D:** the closing tag </note is missing its final '>' character, so the tag cannot be read.

Code (option_a.xml to option_d.xml):

option_a.xml:

```
<note><to>User</to><from>Admin</from>
```

option_b.xml:

```
<note><to>User</to><from>Admin</from></note>
```

option_c.xml:

```
<note><to>User</from></note>
```

option_d.xml:

```
<note to="User"></note
```

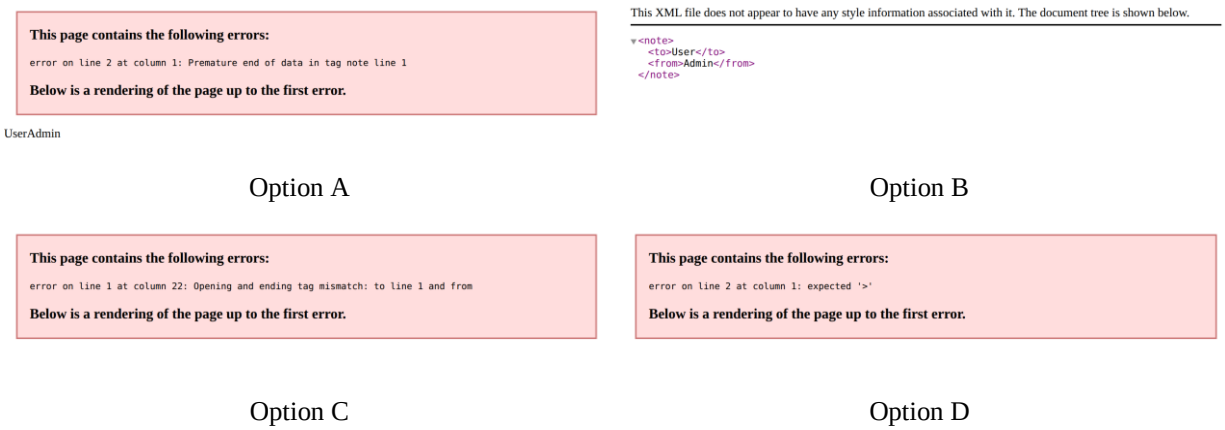


Figure 5: Browser output for each XML option. Only option B loads without an error.

Question 5

Choose the correct protocol for web communication and justify your choice.

Answer: Option B, HTTP (Hypertext Transfer Protocol).

Justification: HTTP is the application-layer protocol that browsers and web servers use to request and deliver web pages, images and form data. Figure 6 shows a real HTTP request sent by a client and the response returned by the server, including the status line 'HTTP/1.0 200 OK'. The other options serve different purposes: FTP transfers files between computers, SMTP sends electronic mail, and SNMP monitors and manages network devices. For production, the web application should use HTTP over TLS (HTTPS) so that communication is encrypted.

```
$ curl -sv http://127.0.0.1:8000/complete.html
> GET /complete.html HTTP/1.1
> Host: 127.0.0.1:8000
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.0 200 OK
< Server: SimpleHTTP/0.6 Python/3.12.3
< Date: Mon, 12 Oct 2026 15:34:11 GMT
< Content-type: text/html
< Content-Length: 1036
< Last-Modified: Mon, 12 Oct 2026 15:34:06 GMT
<
```

HTTP request and response captured with curl

Solution Design: Combined Code-Along Page

The corrected elements are combined into one page. The page contains a semantic navigation menu, the promotional video, and a registration form that submits with POST.

Code (complete.html):

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSA43 - First Web Page</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 20px; }
    nav ul { list-style: none; padding: 0; margin: 0; }
```

```
    nav li { display: inline-block; margin-right: 20px; }
    nav a { text-decoration: none; color: #000; font-weight: bold; }
    nav { border-bottom: 1px solid #000; padding-bottom: 8px; margin-
bottom: 16px; }
    form p { margin: 8px 0; }
  </style>
</head>
<body>
  <nav>
    <ul>
      <li><a href="#">Home</a></li>
      <li><a href="#">Products</a></li>
      <li><a href="#">Contact</a></li>
    </ul>
  </nav>
  <h2>Promotional Video</h2>
  <video src="video.mp4" controls width="400"></video>
  <h2>Register</h2>
  <form action="/submit" method="POST">
    <p>Full name: <input type="text" name="name"></p>
    <p>Email: <input type="email" name="email"></p>
    <p>Password: <input type="password" name="password"></p>
    <button type="submit">Create account</button>
  </form>
</body>
</html>
```

